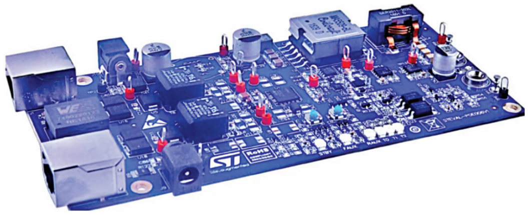
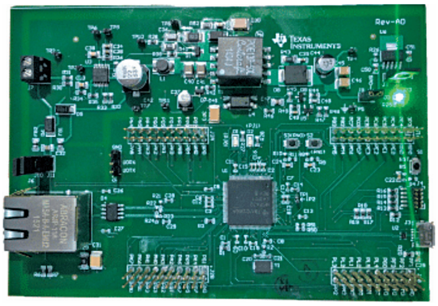
Interesting Reference Designs Of POWER OVER ETHERNET (PoE)

Title of Reference Design	Power Over Ethernet	PoE for Industrial Gateways
Image		
Key Attractions	The design features a 3.3V/20A DC-DC active clamp forward converter with a power over Ethernet interface and dual active bridges. It is suitable for versatile applications, offering operational flexibility and compact integration.	The reference design integrates PoE technology with an Ethernet microcontroller, enabling IoT application development. It supports power extraction from network cabling and facilitates data processing and cloud communication.
Highlights	The reference design from STMicroelectronics, which features a 3.3V/20A DC-DC active clamp forward converter, streamlines the design process for a variety of applications, including wireless access points, and is further enhanced by a power over Ethernet (PoE) PD interface. This design integrates the PM8805 system, which includes two active bridges and an IEEE 802.3bt-compliant powered device (PD) interface, suited for medium-to-high power applications across both 2-pair and 4-pair high-efficiency PoE and PoE+ systems. Equipped with 100V N-channel MOSFETs, the PM8805 offers a low total path resistance of 0.2Ω for each bridge and features capabilities to identify the type of power sourcing equipment (PSE) and provide clear IEEE 802.3af/at/bt classification indications. Additionally, the system offers operational flexibility with smart mode selection, allowing for tailored device functionality, and is packaged in a compact QFN 56 8mm×8mm format for easy integration.	The reference design TIDM-1018 by Texas Instruments integrates its power over Ethernet (PoE) technology with the SimpleLink MSP432E4 Ethernet microcontroller (MCU), enabling the development of Internet of Things (IoT) applications on a compact board. The design allows power to be drawn from existing network cabling while enabling data processing and cloud communication. It includes an RJ45 connector, transformer, and a PoE power stage diode bridge. The board also features a 7W isolated output from a fly-back converter, providing 5V and 3.3V output power rails and an optional power header for external DC power supply in case of network power failure. The board includes dual BoosterPack plug-in module headers compatible with various booster packs for rapid prototyping. The MSP432E401Y MCU operates at 120MHz with 1MB flash and 256kB SRAM, includes cryptographic modules for secure data operations, and additional components.
Applications	The reference design is suitable for compact, high-efficiency applications.	The design is suitable for HVAC gateways, HVAC system controllers, security gateways, and condition monitoring gateways.
OEM Brand	STMicroelectronics	Texas Instruments (TI)
URL	https://www.electronicsforu.com/electronics-projects/power-over-ethernet-reference-design	https://www.electronicsforu.com/electronics-projects/power-over-ethernet-reference-design-for-industrial-gateways